Financial Foundations

Market Microstructure, Venues, and Products Across Asset Classes

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1 Market Microstructure Basics

1.1 The Limit Order Book

Nearly every modern electronic market is organised around a **central limit order book** (CLOB): a price-ordered queue of resting buy orders (the *bid* side) and resting sell orders (the *ask*, or *offer*, side). The book is the venue's state; every message a participant sends either adds to it, removes from it, or crosses it.

Bids		Asks	
Size	Price	Price	Size
500	100.25	100.27	300
800	100.24	100.28	600
1200	100.23	100.29	400
600	100.22	100.30	900

The **best bid** is 100.25 and the **best ask** 100.27, giving a **spread** of \$0.02 and a **mid-price** of 100.26. **Depth** is the quantity resting at each level; **market impact** is the price displacement caused by consuming that depth.

Spread decomposition. The quoted spread compensates a liquidity provider for three distinct costs:

$$\text{Spread} = \underbrace{\text{Order processing}}_{\text{fixed cost per trade}} + \underbrace{\text{Inventory risk}}_{\text{holding an unwanted position}} + \underbrace{\text{Adverse selection}}_{\text{trading against informed flow}}$$

Only the third term is informational, and it is the one that econometric microstructure models try to isolate.

1.2 Matching Algorithms

When an incoming order crosses the book, the exchange's **matching algorithm** decides *which* resting orders at the touched price level get filled. Price priority is universal; venues differ in how they break ties within a price level.

Algorithm	Tie-break rule	Where used
Price–time (FIFO)	Earliest arrival at the price fills first	Equities, most Treasury and equity-index futures (ES, ZN), BrokerTec
Pro rata	Fills allocated proportionally to resting size	Short-term interest rate futures (CME SOFR, Eurodollar legacy)
Modified pro rata	A slice to the price-setter or to time priority, remainder pro rata	CME FX futures, many Eurex contracts
Price–size	Larger resting orders fill first at a price	Some institutional and dealer-to-client platforms
Parity / priority	A fixed percentage to the price-setter, the rest split evenly among quoting market makers	US options exchanges (NYSE American, Arca)

Pro rata worked example (CME SOFR futures). Resting bids at 99.50: A = 1000, B = 500, C = 1500 contracts (total 3000). An incoming sell of 900 contracts is allocated

$$A : 900 \times \tfrac{1000}{3000} = 300, \qquad B : 900 \times \tfrac{500}{3000} = 150, \qquad C : 900 \times \tfrac{1500}{3000} = 450.$$

Because size — not speed — determines the fill, pro rata markets encourage participants to quote far more than they wish to trade. This **size inflation** makes displayed depth a poor proxy for true liquidity, a caveat worth remembering when using book data as a feature.

1.3 Order Types

Order type	Behaviour
Market	Execute immediately against the book at whatever prices are available.
Limit	Execute only at the stated price or better; rests in the book otherwise.
Stop (stop-market)	Becomes a market order once a trigger price trades.
Stop-limit	Becomes a limit order once the trigger price trades.
IOC	Immediate-or-cancel: fill what you can now, cancel the rest.
FOK	Fill-or-kill: fill the entire quantity immediately or cancel it all.
AON	All-or-none: no partial fills, but may rest in the book.
Day / GTC / GTD	Time-in-force: expires at the close / good till cancelled / good till date.
Iceberg (reserve)	Displays a small tip; the hidden remainder replenishes as the tip is hit, usually losing time priority on each refresh.
Pegged	Price tracks a reference (mid, primary bid, NBBO) as the reference moves.
Hidden	Fully non-displayed; typically yields priority to displayed orders.
MOO / MOC / LOC	Participates only in the opening or closing auction.

1.4 Major Electronic Trading Platforms

The transition from open outcry to electronic matching (NASDAQ electronic from 1971; NYSE and the CME/CBOT pits substantially electronic by the mid-2000s) compressed execution latency from minutes to microseconds and replaced designated floor intermediaries with algorithmic liquidity provision. The table below lists the venues that matter for the data we will use in this course.

Asset class	Platform	Notes
Equities (US)	NYSE, NYSE Arca, American, National, Texas	Auction open/close; DMM model on the NYSE floor
	Nasdaq, BX, PSX	Fully electronic since inception
	Cboe BZX, BYX, EDGA, EDGX	Four books with distinct fee/rebate models
	IEX, MEMX, MIAX Pearl, LTSE	Newer entrants; IEX runs a 350 μs speed bump
	30+ ATSs / dark pools	Non-displayed; report to TRF, print to the tape
Equities (non-US)	LSE, Euronext, Deutsche Börse Xetra, SIX	European primary listing venues
	Cboe Europe, Turquoise, Aquis	European MTFs competing with the primaries
	JPX, HKEX, SSE/SZSE, KRX, NSE/BSE, ASX, TSX, B3	Asian and other regional primaries
Commodity futures	CME Globex (CME, CBOT, NYMEX, COMEX)	Energy, metals, grains, livestock; the deepest book
	ICE Futures Europe / US / Endex	Brent, gas oil, TTF gas, sugar, coffee, cocoa, cotton
	LME Select / LMEselect + Ring	Base metals; unique prompt-date structure
	Euronext, SGX, DME, MGEX, DCE/ZCE/SHFE/INE	Regional grains, iron ore, Oman crude, Chinese commodity complex
FX	EBS Market (CME)	Primary interbank CLOB: EUR/USD, USD/JPY, USD/CHF
	LSEG (Refinitiv) Matching / FXall	Primary CLOB for GBP, CAD, Scandies; RFQ for the buy side
	Cboe FX (Hotspot), Currenex, 360T, FXSpotStream	ECNs and aggregators
	CME, ICE, Eurex	Listed FX futures and options
Fixed income	BrokerTec (CME)	Interdealer Treasuries and repo; anonymous, price–time
	Fenics UST, Dealerweb	Competing interdealer Treasury books
	Tradeweb, Bloomberg (TSOX/ALLQ)	Dealer-to-client RFQ, governments and credit
	MarketAxess	Corporate bonds; all-to-all "Open Trading" protocol

1.5 Special Order Types

Several widely used order types are defined by reference to a price that is not yet known at the time of the trade — most often the exchange's **official settlement price**, a volume-weighted average struck over a short closing window rather than the last trade.

Order type	Purpose
End-of-day (EOD)	Executes at or near the settlement price, minimising tracking error against a benchmark struck at the close.
Trade at settlement (TAS)	Trades at the official settlement price, or at a small tick offset to it, with the price unknown at trade time. Widely used in commodity futures for index replication and portfolio rebalancing.
Trade at marker (TAM)	As TAS, but referenced to an intraday marker price rather than the daily settle.
Basis trade at index close (BTIC)	Equity-index futures traded at a spread to that day's official cash index close.
Exchange for physical (EFP)	Privately negotiated swap of a futures position for the equivalent cash position; reported to the exchange, not matched on the book.
Block trade	Large privately negotiated trade above a minimum size threshold, executed away from the book and reported with a delay.
Implied pricing	The exchange (CME, ICE) synthesises orders in calendar spreads and combinations from outrights (*implied in*) and outrights from spreads (*implied out*), tightening spreads and linking liquidity across the curve. The implied book is real, tradeable liquidity that does not appear in the outright order flow.

2 Equities

2.1 Main Trading Venues

The US equity market is the most fragmented in the world: roughly 16 registered national securities exchanges plus more than 30 alternative trading systems (ATSs), all trading the same securities simultaneously. Regulation NMS binds them together.

Reg NMS and the NBBO. The **national best bid and offer** is the consolidated best displayed price across all protected venues. The order protection rule forbids executing at a price inferior to a protected quote elsewhere, which forces brokers to route across venues and makes **consolidated tape** (SIP) data — as distinct from single-venue direct feeds — the reference for best execution. Direct feeds are faster; the SIP is authoritative. Latency arbitrage between the two is a first-order microstructure phenomenon.

2.2 Main Products

A **share** of common stock is a unit of ownership in a corporation, conveying a *residual* claim — on cash flows and, in liquidation, on assets — that ranks behind every other claimant: secured and unsecured creditors, then preferred stock.

- **Common and preferred stock**, and **ADRs** for foreign issuers.
- **ETFs**, whose creation/redemption mechanism ties the traded price to NAV via authorised participants; ETF arbitrage is one of the cleanest statistical relationships in equity markets.
- **Listed equity and index options**, traded across 16 US options exchanges.
- **Equity-index futures** (ES, NQ, RTY and their micro variants) on CME Globex, which often lead the cash market in price discovery.

Options exchange	Specialisation	Feature
Cboe	Index options, VIX	Largest US options venue; SPX and VIX complex
Nasdaq PHLX	Equity and index options	Maker/taker pricing
NYSE American	Equity options	Price improvement auction
NYSE Arca	ETF and equity options	Parity/priority allocation
BOX	Equity options	Price improvement period (PIP)
MIAX, MIAX Pearl, Emerald	Equity options	Proprietary matching and auction mechanisms

Contract conventions. Standard monthlies expire the third Friday; single-name options are American-style with physical settlement, while broad index options are European-style and cash-settled. Assignment is automatic if in the money by \$0.01. Beyond the monthlies: **weeklies** (every Friday), **end-of-month** options, **quarterlies**, **LEAPS** (out to three years), and **0DTE** contracts, which now expire every trading day on SPX and the largest ETFs and account for a large share of SPX option volume.

2.3 Special Features

Auctions.
The opening and closing auctions concentrate enormous volume into single events. The closing auction sets the official close used by index providers and mutual funds, and has grown to roughly a tenth of daily consolidated volume.

Corporate actions.
Splits, dividends, spin-offs, mergers, and rights issues all require price and volume series to be back-adjusted. Failing to adjust produces spurious returns; this is the single most common data-quality failure in equity research.

Short selling.
Locate requirements, borrow costs, bi-monthly short interest reporting, and days-to-cover ratios all matter for strategies with a short leg, and borrow cost is a persistent, forecastable drag.

3 Commodity Futures

3.1 Main Trading Venues

Commodity trading is concentrated in a small number of futures exchanges, each effectively holding a franchise in particular contracts. Unlike equities, there is no fragmentation: liquidity in WTI crude lives on NYMEX and essentially nowhere else, so a single venue's book is the market.

Exchange	Franchise contracts
NYMEX (CME)	WTI crude (CL), Henry Hub natural gas (NG), RBOB gasoline (RB), heating oil / ULSD (HO)
COMEX (CME)	Gold (GC), silver (SI), copper (HG)
CBOT (CME)	Corn (ZC), soybeans (ZS), wheat (ZW), soybean oil and meal
CME	Live cattle (LE), lean hogs (HE), dairy
ICE Futures Europe	Brent crude (B), gas oil, TTF and NBP natural gas
ICE Futures US	Sugar #11, coffee C, cocoa, cotton #2, FCOJ
LME	Aluminium, copper, zinc, nickel, lead, tin
Others	Euronext milling wheat/rapeseed, SGX iron ore, DME Oman crude, MGEX spring wheat, DCE/ZCE/SHFE/INE

3.2 Main Products

A **futures contract** is a standardised, exchange-listed agreement obligating one party to buy and the other to sell a specified quantity and grade of an underlying at a price fixed today, for settlement on a specified future date. Around it sit options on futures, calendar spreads, crack and crush spreads, and cash-settled index products.

The rolling problem. Every futures contract expires, so a continuous price history must be constructed by splicing contracts together. The splice introduces an artificial jump equal to the inter-contract price difference. Standard remedies are **back-adjustment** (shift the history by the roll gap, preserving returns but distorting levels, possibly into negative prices) or **ratio adjustment** (scale by the price ratio). There is no choice that preserves both levels and returns; the roll convention must be documented alongside any backtest.

3.3 Special Features

Physical delivery and storage.
Contracts are anchored to warehouses, pipelines, and delivery points. Storage costs, delivery logistics, and grade differentials all enter the price, and the LME's prompt-date structure and warehouse queues are a market unto themselves.

Term structure.
Contango ($F_t > S_t$) prevails when storage costs dominate; **backwardation** ($F_t < S_t$) when the *convenience yield* of holding the physical dominates. The resulting **roll return** between consecutive expiries $T_1 < T_2$ is

$$\text{Roll return} = \frac{F_{t,T_1} - F_{t,T_2}}{F_{t,T_2}},$$

positive in backwardation and negative in contango. For a passive long index position the roll return, not the spot move, often dominates total return.

Position limits and price limits.
The CFTC and exchanges impose speculative position limits and daily price limits (limit-up/limit-down), which can halt trading and truncate the observed return distribution on exactly the days a risk model cares most about.

Commitments of Traders.
The weekly CFTC report decomposes open interest into commercial (hedgers), non-commercial (speculators), and non-reportable categories, with concentration ratios for the four and eight largest traders. It is one of the few public positioning datasets in any market.

4 Foreign Exchange

4.1 Main Trading Venues

FX market characteristics.

- Around 90% of spot FX trades over the counter, decentralised across many competing liquidity pools rather than on a single exchange.
- Trading runs 24 hours a day, five days a week, rolling through the Sydney, Tokyo, London, and New York sessions.

Venue	Segment	Role
EBS Market (CME)	Interbank spot	Primary CLOB for EUR/USD, USD/JPY, USD/CHF
LSEG (Refinitiv) Matching	Interbank spot	Primary CLOB for GBP, CAD, AUD, Scandinavians
LSEG FXall	Dealer-to-client	RFQ across spot, forwards, swaps, options
Cboe FX (Hotspot), Currenex	ECN	Anonymous all-to-all pools
360T, FXSpotStream	Aggregator	Multi-dealer streaming and RFQ
Single-dealer platforms	Principal	Bank-internalised risk transfer
CME, ICE, Eurex	Listed	FX futures and options on futures
CLS	Settlement	Payment-versus-payment for 18 currencies

4.2 Main Products

Instrument	Definition
Spot	Exchange at the current rate, settling T+2 (T+1 for USD/CAD).
Outright forward	Exchange at an agreed rate on a customised future date.
FX swap	Simultaneous spot purchase and forward sale (or vice versa); the dominant instrument by turnover, used for funding rather than directional risk.
NDF	Non-deliverable forward, cash-settled in USD against a fixing; used for restricted currencies (KRW, TWD, INR, BRL).
Cross-currency basis swap	Exchange of interest streams and principal in two currencies.
Options	Vanilla European/American plus a large exotic market (barriers, digitals).
Futures	Listed, centrally cleared, quoted in the reciprocal convention to spot.

Pairs are grouped into **majors** (EUR/USD, USD/JPY, GBP/USD, USD/CHF, AUD/USD, USD/CAD, NZD/USD), **crosses** (pairs without USD, e.g. EUR/JPY), and **emerging markets**. Quotes are given as units of the quote currency per unit of base currency, with the **pip** the fourth decimal place for most pairs and the second for JPY pairs.

4.3 Special Features

No consolidated tape.
Because there is no single venue, "the" FX price does not exist. Different data vendors will disagree, and any research result must state which feed it used.

Last look.
Many liquidity providers reserve a brief window to reject a trade after it is requested. Fill rates and rejection asymmetry are therefore part of the true cost of trading, and quoted spreads systematically understate that cost.

Benchmark fixes.
The WM/Refinitiv 4pm London fix is the benchmark for index and portfolio valuation, and concentrates a large volume spike into a short window each day.

Value dates and rolls.
Spot positions held past the cut roll to the next value date via tom/next swaps, so the carry accrues through the swap points rather than through the spot rate.

Triangular consistency.
Cross rates are pinned to the majors by arbitrage (EUR/JPY = EUR/USD × USD/JPY), which makes FX one of the few markets with an exact, testable no-arbitrage relationship at high frequency.

Carry and interest rate parity.
Forward points are set by the interest rate differential via covered interest parity; the systematic failure of *uncovered* parity is the basis of the carry trade, whose returns are negatively skewed and crash-prone.

5 Fixed Income

5.1 Main Trading Venues

Fixed income remains a predominantly dealer-intermediated, over-the-counter market, though the most liquid segments have migrated to electronic order books.

Venue	Segment	Protocol
Treasury auctions (TreasuryDirect)	Primary	Uniform-price competitive auction via primary dealers
BrokerTec (CME)	Interdealer USTs, repo	Anonymous CLOB, price–time priority
Fenics UST, Dealerweb	Interdealer USTs	Competing CLOBs
Tradeweb	Dealer-to-client	RFQ across governments, credit, mortgages
Bloomberg TSOX / ALLQ	Dealer-to-client	Terminal-integrated RFQ and streaming
MarketAxess	Corporates	All-to-all "Open Trading" plus classic RFQ
CME (CBOT)	Listed rates	Treasury and SOFR futures and options

Post-trade transparency comes from **TRACE** for corporates and **EMMA/MSRB** for municipals; there is nothing comparable to an equity consolidated tape for pre-trade quotes.

5.2 Main Products

- **US Treasuries:** bills (discount, $\leq$ 1y), notes (2–10y), bonds (20–30y), TIPS, and floating rate notes.
- **Agency MBS:** traded largely in the **TBA** market, where the specific pools are not identified until settlement.
- **Corporate bonds:** investment grade and high yield, with tens of thousands of outstanding CUSIPs, most of which do not trade on a given day.
- **Municipals:** tax-exempt state and local debt, extremely fragmented.
- **Repo:** the collateralised funding market that underpins dealer inventory and is the source of the SOFR benchmark.
- **Listed rate futures:** ZT, ZF, ZN, TN, ZB, UB on the Treasury curve, and SOFR futures (SR1, SR3) on the short end.

5.3 Special Features

Yield-space pricing.
Bonds are quoted and analysed in yield, not price. The mapping is non-linear, so returns are modelled through **duration** and **convexity**, and risk through **DV01**, the price change per basis point. Treasury prices are quoted in thirty-seconds of a point.

Clean versus dirty price.
The quoted (clean) price excludes accrued interest; the invoice (dirty) price includes it. Day-count conventions (actual/actual for Treasuries, 30/360 for corporates) differ by instrument and must be applied correctly or accrual-driven sawtooth artefacts will appear in any price series.

On-the-run versus off-the-run.
The most recently auctioned issue at each tenor concentrates nearly all liquidity and trades at a premium. Yield curve construction must decide explicitly which issues to use, and the on-the-run/off-the-run spread is itself a tradeable liquidity signal.

Sparse trading.
Because most bonds are held to maturity, "last price" for a corporate bond may be days old. Evaluated pricing services interpolate, which means much apparent bond price data is model output rather than observed transactions.

Cheapest to deliver.
Treasury futures permit delivery of any bond in a basket; the **cheapest-to-deliver** issue and its conversion factor determine the futures price and make the cash–futures basis a distinct traded market.

Matching algorithm.
SOFR futures use pro rata allocation, unlike the price–time priority of Treasury futures — a direct consequence of the very tight tick relative to typical order size at the short end of the curve.